

Twitter announces the Heron open source stream processing engine

Twitter has released the code of Heron, a stream-processing engine that succeeded Apache Storm.



Licensed under Apache v2.0, the distributed stream computation system has powered real-time analytics at Twitter for the past two years.

Heron was born out of the challenges that Twitter faced with the increase in the volume and diversity of data and number of use cases. The internal team decided to develop a streaming system that would be easy to scale, debug, deploy and manage after encountering numerous problems with the previously deployed Storm.

"Heron represents a fundamental change in streaming architecture from a thread-based system to a process-based system," said Karthik Ramasamy, engineering manager at Twitter, in a blog post.

Some startups have deployed Heron on their projects. Even Microsoft is using a modified version of Heron that runs on top of a YARN cluster management component of Hadoop. The technology supports use cases ranging from extract transform load (ETL) to advertising bidding and even augmented reality (AR), up to a certain level.

Heron is written in languages like Java, C++ and Python to deliver efficiency, maintainability and easier adoption from the open source community. The system supports modern cluster environments such as Apache Mesos, Aurora and REEF, among others.

In addition to Heron, Twitter has open sourced a couple of more major projects like Diffy, Scalding and Summingbird. All these projects are live on GitHub.

Ansible is well established among businesses. In fact, a large number of Fortune 100 companies are already using Ansible automation to maintain their private cloud environments. Ansible Roles works as pre-packaged, ready-to-run sys admin tasks to provide developers a simple way to access their servers.

However, the open sourcing of the Ansible project helps companies set up a private Galaxy server and redirect it to their private Galaxy repository. There are also options to let developers and systems administrators submit new features and enhancements directly to the Ansible Galaxy codebase using the open source solution.

Software developers can access the Ansible project using their existing GitHub account. The Python based code is also available on the online repository to customise experience.

Facebook develops open source Javascript manager Yarn



Facebook has released its open source JavaScript package manager called Yarn. This new development is supposed to offer faster and more reliable installs than the original npm and is compatible with the npm registry.

According to an official statement by Facebook, Yarn is capable of reducing install times from many minutes to just a few seconds. The package manager uses lockfiles and a deterministic installation algorithm to deliver efficient results. Also, it eases the process of maintaining `node_modules` directories.

"Yarn is a new package manager that replaces the existing workflow for the npm client or other package managers while remaining compatible with the npm registry," the Facebook engineering team led by Sebastian McKenzie, Christoph Pojer and James Kyle wrote in a recent post.

By caching all newly downloaded packages, Yarn ensures quicker and reliable installs. This also helps in installing the packages even when you don't have an active Internet connection. However, you need to have installed the same package at least once to cache the data. Facebook confirms that Yarn is compatible with both npm and Bower workflows.

Yarn comes months after the social networking company faced numerous issues with npm. It was using the default JavaScript package manager internally for a long time. And finally, Yarn emerged through a partnership between Facebook engineers and multiple teams of developers from Google, Exponent and Tilde.

You can experience the advantages of Yarn over the original npm by downloading its package from its official website. Its code is also available openly on GitHub.

Cyanogen picks modular programme to 'do more' with Android

Cyanogen has finally shifted its focus from a complete custom Android platform and announced a new modular programme. The company has also reshuffled its leadership by transitioning its co-founder and CEO, Kirt McMaster, into an executive chairman role and selecting former COO Lior Tal as the new CEO.

Tal will focus the company's efforts into making the new programme successful. Additionally, Steve Kondik, co-founder and present CTO of Cyanogen, will be taking on a new role as chief science officer. Kondik will report to Stephen Lawler, SVP of engineering at Cyanogen.